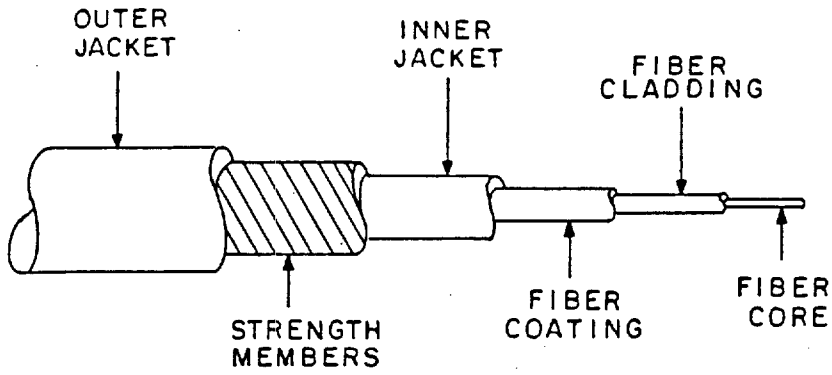
MILITARY SPECIFICATION SHEET

CABLES, FIBER OPTIC, HIGH RELIABILITY, SINGLE FIBER, MULTI-MODE, GRADED INDEX, OUTGASSING RESISTANT

This specification is approved for use by the Aeronautics and Space Administration and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: DOD-C-85045.



Part or identifying number (PIN)	Inner jacket (buffer tube) diameter inches and (mm)			Outer Jacket diameter inches and (mm)	
	ID	OD	Type	ID	OD
M85045/12C1A	.029 ±.003 (0.74 ±0.08)	.059 ±.003 (1.50 ±0.08)	Loose	.100 ±.005 (2.5 ±0.13)	.120 ±.002 (3.0 ±0.05)

FIGURE 1. Dimensions and configuration.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Weight: 10 grams (0.36 oz.) per meter.

Fiber: Type I, class 1, composition A, style 2, 100 µm ±3 µm core, 140 µm ±2 µm cladding, in accordance with DOD-F-49291. Fiber coating material; Acrylate polymer.

Number of fibers: One.

Jacket material: Ethylene - tetrafluoroethylene polymer.

Inner jacket (buffer tube) material: Polyester elastomer.

Filler and strength member material: Fiberglass - polytetrafluoroethylene braid.

Color: Outer cable jacket shall be black.

Attenuation: 5 dB/km maximum at 850 nm ±30 nm.

Optical fiber bandwidth: 100 MHz-km minimum at 850 nm ±30 nm.

Numerical aperture: 0.29 ±.02.

Temperature range:

Operating: -46°C to +71°C

Storage: -55°C to +85°C

Temperature cycling: After being subjected to 5 cycles of temperature cycling, the induced attenuation shall not exceed 1 dB. Temperatures are -46°C +2°C and +71°C +2°C. A single chamber shall be used and the rate of temperature change shall be 10°C/min over the rated temperature range.

Humidity: 85 percent at +85°C for 240 hours.

Barometric pressure (reduced): 1×10^{-6} atmosphere at +71°C ±2°C.

Life at elevated temperature: 2,000 hours at +71°C. 0.5 dB allowable change in attenuation.

Operating tensile load: Applicable.

Fungus: Not applicable.

Low temperature flex: Not applicable.

Cyclic flexing: Not applicable.

Crush: 200 N load, 25 mm OD mandrel.

Cable twist-bend: Cable attenuation shall not increase by more than 0.5 dB when tested in accordance with EIA-455-91, 180° bend, 25 mm OD mandrel, and a 10 N load. 2,000 cycles duration.

Impact: 150 mm height, 12.5 mm OD mandrel, 0.5 kg load.

Jacket self adhesion or blocking: Not applicable.

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Wicking: Not applicable.

Jacket opacity: Not applicable.

Fluid immersion: Not applicable.

Corner bend: Not applicable.

Ultraviolet radiation: Not applicable.

Flammability: 60° angle procedure. After flame duration - 10 seconds. After flame travel - 1 cm.

Vibration: Cable attenuation shall not increase by more than 0.5 dB when subjected to a vibration test, at 20 to 2,000 Hz in accordance with EIA-455-11. Peak acceleration shall be 20 g. Cable shall pass visual examination after the test.

Shock: Cable attenuation shall not increase by more than 0.5 dB when subjected to the shock test in accordance with EIA-455-14, condition A and as specified herein. Peak acceleration shall be 50 g. Shock shall be applied perpendicular to the light transmission axis. Cables shall pass visual examination after the test.

Thermal stability: Cable attenuation, with reference to the +25°C levels, shall not change by more than ±1 dB when measured at +20°C increments over the operating temperature range of -46°C to +71°C, when tested in accordance with DOD-STD-1678 method 4010.

Vacuum effects, outgassing: Total mass loss (TML) shall not exceed 1 percent and the collected volatile condensable material (CVCM) shall not exceed 0.1 percent when tested in accordance with ASTM E-595.

Nuclear radiation: Radiation test and measurements shall be performed at +25°C ±5°C in accordance with DOD-F-49291 appendix except that the source and launch conditions shall be as specified in EIA-455-53. Cable attenuation shall not increase by more than 1 dB for a 10 meter length of cable, at radiation total dose levels of 10 kilorads (Si) at 50 rads minimum dose rate. Cable attenuation shall return to the preradiation values (room temperature anneal) within ±0.5 dB, 24 hours after radiation.

Qualification inspection shall consist of performing the inspections and optical tests specified in table I herein, in the sequence shown.

Quality conformance inspection shall be performed as specified in DOD-C-85045 and as specified herein.

Group A inspection shall consist of performing the inspections and optical test specified in table II herein, in the sequence shown. The acceptable quality level (AQC) shall be 1 percent major, 4 percent minor.

Group B inspection shall consist of performing the inspections and optical test specified in table III herein, in the sequence shown. The acceptable quality level (AQC) shall be 1 percent major, 4 percent minor.

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TABLE I. Qualification inspection.

Inspection	Sample quantity	Sample length
Group IA 1/		
Visual and mechanical examination	100%	
Attenuation	100%	
Bandwidth	2 units	100 m
Numerical aperture	2 units	100 m
Thermal stability	2 units	10 m
Group IB		
Temperature cycling	2 units	10 m
Nuclear radiation	2 units	10 m
Life at elevated temperature	10 units	10 m
Group II		
Barometric pressure (reduced)	2 units	10 m
Humidity	2 units	10 m
Cable twist-bend	2 units	10 m
Group III		
Vibration	2 units	10 m
Shock	2 units	10 m
Tensile load	2 units	10 m
Impact	2 units	10 m
Crush	2 units	10 m
Group IV 2/		
Flammability	2 units	1 m
Vacuum effects, outgassing	2 units	1 m
Cable element, removability	2 units	1 m

1/ Groups IB through III samples shall be taken (and terminated) from group IA.

2/ Separate, unterminated segments of sample cables from group IA shall be assigned to each group IV test.

TABLE II. Group A inspection (100%).

Inspection or test
Visual and mechanical
Attenuation

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TABLE III. Group B inspection. 1/

Inspection	Sample quantity	Sample length
Group I		
Bandwidth	2 units	100 m
Numerical aperture	2 units	10 m
Group II		
Temperature cycling	4 units	10 m
Thermal stability	2 units	10 m
Group III		
Vibration	2 units	10 m
Tensile load	2 units	10 m

1/ Group B test units shall have passed group A inspection on table II.

Intended use: Cables specified in this document are high reliability fiber optic cables which exhibit excellent materials outgassing characteristics and are intended for use in space flight and critical ground support equipment applications.

Part or identifying number: M85045/12-C1A.

CONCLUDING MATERIAL

Custodian:
NASA - NA

Review activity:
DLA - ES

Preparing activity:
NASA - NA

Agent:
DLA - ES

(Project 6015-K013)

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